CONFOEDERATIO,
RESEARCH DIVISION (CRD).

THE PRINCIPALITIES OF CHERNIGOV AND NOVGOROD-SEVERSK

A Regional Literature Review of Population and Settlement (10000BC - 1300AD).

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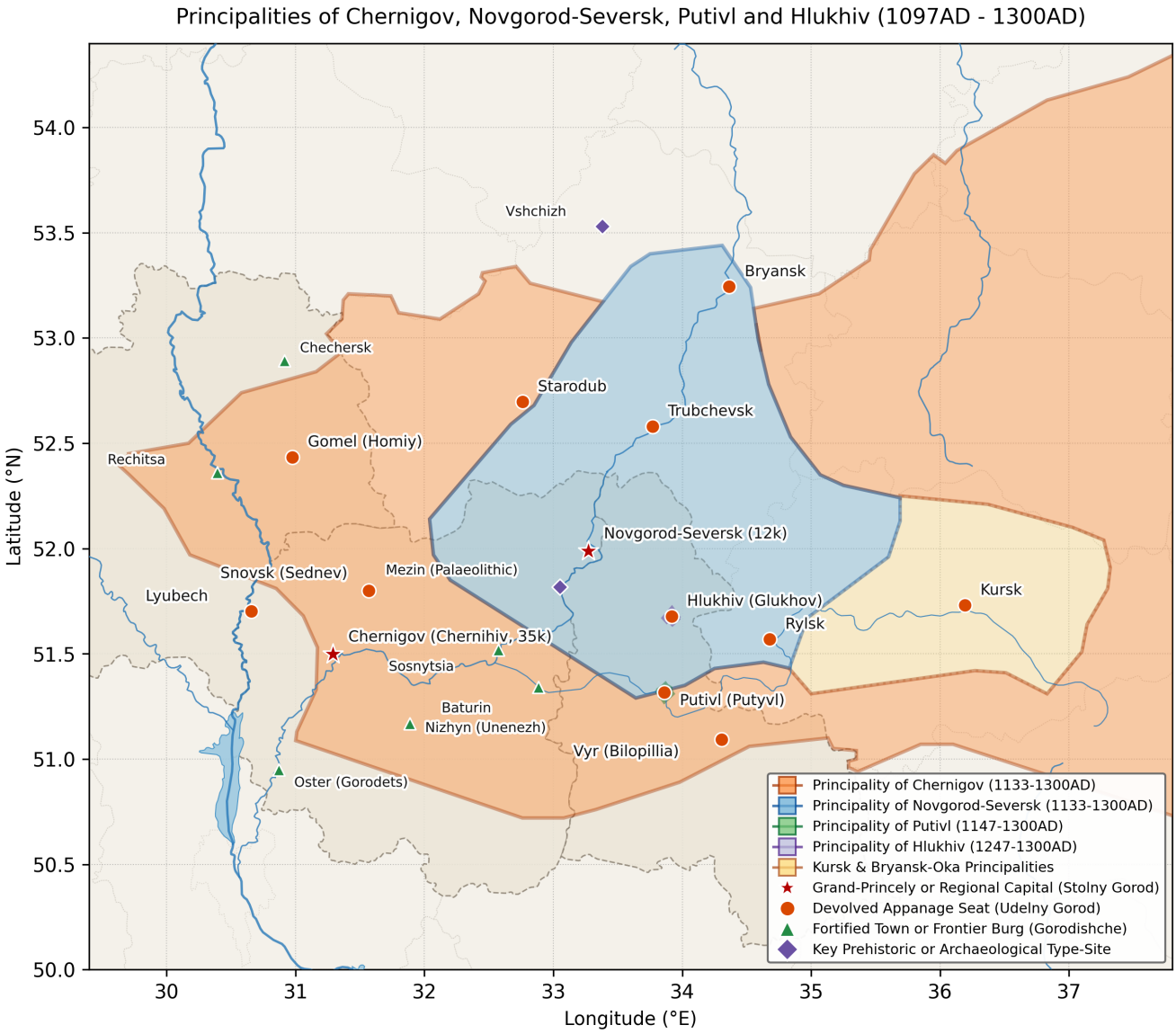


Figure 1. Regional boundaries and fortified settlement hierarchy of the principalities of Chernigov, Novgorod-Seversk, Putivl, Vorgol, and Hlukhiv.

Executive Summary and Spatial Definition

Scope of the Regional Synthesis (10000BC - 1300AD)

This monograph establishes the long-run prehistoric, proto-Slavic, Kievan Rus, and Mongol-conquest demographic trajectory (10000BC - 1300AD) alongside the settlement-level archaeological and chronicle synthesis for the Left-Bank Desna, Seym, Snov, and Lower Sozh basins (Chernigov and Novgorod-Seversk). During the eleventh to thirteenth centuries, the wider Chernigov-Seversk realm ruled by the Olgovichi branch of the Rurikid dynasty constituted the largest territorial conglomerate in Eastern Europe, stretching from the Dnieper-Desna-Sozh river junctions in the southwest to the Upper Oka, Moscow river approaches, and Kursk Posemye in the northeast [1][2][3].

To maintain strict spatial non-overlap across the macro-regional demographic series, the wider Chernigov-Seversk realm is partitioned along modern administrative boundaries into two complementary analytical zones. The Southern Core of Chernigov and Novgorod-Seversk (72 235 km^2) encompasses the historic dynastic capitals and southern forest-steppe heartland across Chernihivska Oblast (Ukraine, 31 903 km^2), Sumska Oblast (Ukraine, 23 832 km^2), and the Eastern Gomel and Lower Sozh-Dnieper Corridor (Belarus, 16 500 km^2). This territory contains the primary territorial domains of the Principality of Chernigov, the Principality of Novgorod-Seversk, the Principality of Putivl, the Principality of Vorgol, and the post-Mongol Principality of Hlukhiv. The Northern Russian Half (~118 000 km^2) encompasses the northern forest and north-eastern forest-steppe appanages inside the modern Russian Federation, accounted for separately in the Russian Core synthesis across the Upper Oka and Bryansk-Chernigov North (Bryansk, Starodub, Trubchevsk, Vshchizh, Karachev, Novosil, Tarusa, Mtsensk, Odoev, Vorotynsk, Belyov, and Kozelsk) and the Kursk-Seversk Steppe Frontier (Kursk, Rylsk, and Yelets).

At the 1200AD pre-Mongol apogee, the Southern Core reconstructed in this study supported 460 000 inhabitants (360 000 - 560 000, mean density 6,37 inh/km^2), anchored by the metropolitan complex of Chernigov (35 000 inhabitants across 330 ha) and the princely capitals of Novgorod-Seversk (12 000), Putivl (7 500), Gomel (5 000), and Hlukhiv (5 000) [4][5][6]. When combined with the 500 000 inhabitants reconstructed for the Northern Russian Half (210 000 in Bryansk-Starodub-Upper Oka plus 290 000 in Kursk-Rylsk Posemye), the Grand Principality of Chernigov-Seversk as a whole contained 960 000 inhabitants at 1200AD, rivalled within the Rus lands only by the combined Kievan-Galician-Volhynian south [7]. Following the catastrophic October 1239AD Mongol invasion under Mengu-Timur, the Southern Core contracted by 54,3% to 210 000 in 1250AD before recovering partially to 260 000 by 1300AD under the sheltered post-conquest principalities of Hlukhiv, Putivl, and post-Mongol Chernigov.

Summary Benchmark Table (10000BC - 1300AD)

Table 1 summarises the reconstructed population trajectory for the Southern Core (72 235 km^2) across sixteen benchmark horizons from the Late Pleniglacial and Bølling-Allerød mammoth-hunter complexes to the post-Mongol devolved principalities at 1300AD.

Epoch	Best Estimate	Uncertainty Band	Density (inh/km2)	Urban Share (%)	Primary Evidentiary Basis
10000BC	4 500	2 500 - 7 500	0,06	0,0%	Mezin, Pushkari I & Chulatovo mammoth-bone camps
6000BC	9 000	5 500 - 14 000	0,12	0,0%	Desna-Sozh Mesolithic & early Pit-Comb Ware sites
4000BC	18 000	11 000 - 27 000	0,25	0,0%	Dnieper-Donets II & Desna Neolithic fishers

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Epoch	Best Estimate	Uncertainty Band	Density (inh/km2)	Urban Share (%)	Primary Evidentiary Basis
1500BC	52 000	34 000 - 74 000	0,72	0,0%	Sosnytsia & Marianivka Bronze Age agro-pastoralists
500BC	85 000	55 000 - 120 000	1,18	0,0%	Yukhnovo, Milograd & Scythian Posemye hillforts
0AD	105 000	70 000 - 145 000	1,45	0,0%	Late Zarubintsy (Pochep-Harivka) & Late Scythian
300AD	140 000	95 000 - 190 000	1,94	0,0%	Kiev culture (Roishche) & northern Chernyakhov peak
500AD	115 000	75 000 - 160 000	1,59	0,0%	Kolochin culture forest-riverine refugia
700AD	165 000	115 000 - 220 000	2,28	2,5%	Volyntsevo & early Romny-Borshchevo Severian opoliya
800AD	190 000	135 000 - 250 000	2,63	4,8%	Bititsa, Gornal & Romny fortified Severian hillforts
900AD	220 000	160 000 - 285 000	3,05	8,5%	Shestovytsia, Chernigov barrows & Lyubech trade axis
1000AD	280 000	210 000 - 355 000	3,88	13,2%	Early Rurikid tribute network & Severian integration
1100AD	365 000	285 000 - 450 000	5,05	17,8%	Post-1097AD Lyubech Congress Olgovichi expansion
1200AD	460 000	360 000 - 560 000	6,37	21,5%	Olgovichi apogee (116 hillforts; Chernigov 35k)
1250AD	210 000	155 000 - 275 000	2,91	10,5%	October 1239AD Mongol destruction of Desna-Posemye
1300AD	260 000	195 000 - 330 000	3,60	11,8%	Post-Mongol Hlukhiv, Putivl & Chernigov principalities

Table 1. Reconstructed benchmark population, settlement density, and urban share for the southern Chernigov-Seversk core (10000BC - 1300AD).

Devolved Principalities and Areal Masks

Devolved Principalities and Historical Boundaries (1097AD - 1300AD)

The political geography of the Left-Bank Desna, Seym, Snov, and Lower Sozh basin reflects the progressive dynastic ramification of the Olgovichi (descendants of Oleg Svyatoslavich, d. 1115AD, who secured hereditary rights to the Chernigov land at the Congress of Lyubech in 1097AD) [2][3]. Following Vsevolod Olgovich's accession to the Kievan throne (1139AD), the realm divided into the senior Chernigov patrimony (*Otchina*) along the Lower and Middle Desna, Snov, and Sozh, and the junior Seversk (*Sever*) patrimony along the Middle Desna and Seym. After the Mongol destruction of 1239AD and the martyrdom of Grand Prince Mikhail Vsevolodovich at the Golden Horde (1246AD), the southern lands reorganised into the post-Mongol devolved principalities of Chernigov, Novgorod-Seversk, Putivl, and Hlukhiv (founded 1247AD). Table 2 details the devolved principalities represented within the Southern Core territory.

Dynastic Phase	Principality Name	Active Span	Capital (Coords)	Core Basin & Administrative Coverage
Senior Patrimony	Principality of Chernigov	1133-1279AD	Chernigov (51,50°N, 31,29°E)	Lower & Middle Desna, Snov, Oster & Lower Sozh (Chernihiv & Gomel)
Post-Mongol Core	Principality of Chernigov	1279-1356AD	Chernigov (51,50°N, 31,29°E)	Post-Mongol Desna-Snov-Sozh core under Bryansk-Chernigov line
Late Appanage	Principality of Chernigov	1356-1358AD	Chernigov (51,50°N, 31,29°E)	Late post-Mongol Chernigov domain prior to Lithuanian annexation
Seversk Patrimony	Principality of Novgorod-Seversk	1133-1363AD	Novgorod-Seversk (51,99°N, 33,27°E)	Middle Desna, Lower Seym & Ivot-Esman (NE Chernihiv & N Sumy)
Posemye Appanage	Principality of Putivl	1147-1501AD	Putivl (51,32°N, 33,86°E)	Middle Seym (Posemye), Vyr, Kleven & Upper Sula & Psyol (Sumy)
Kleven Appanage	Principality of Vorgol	1203-1241AD	Vorgol (51,48°N, 33,78°E)	Kleven-Seym interfluve NW of Putivl (Krolevets & Putyvl)
Esman-Kleven Refuge	Principality of Hlukhiv	1247-1408AD	Hlukhiv (51,68°N, 33,92°E)	Post-Mongol Esman-Kleven plateau (Hlukhiv & Yampil)

Table 2. Devolved Rurikid principalities, active chronological spans, capitals, and river basins across the southern Chernigov-Seversk lands.

Relation to the Northern Desna, Upper Oka, and Kursk Appanages

Because historical and archaeological monographs by A. K. Zaitsev [1], V. P. Kovalenko [4], A. V. Kuza [6][8], and M. Dimnik [3] frequently cite aggregate figures for the entire Chernigov-Seversk realm from the Desna mouth to the Upper Oka and Kursk Posemye, explicit accounting is required to link this Southern Core synthesis with the Northern Russian appanages documented in the European Russia regional review.

In the northern and north-eastern wings of the Olgovichi patrimony, the Upper Oka and Bryansk-Chernigov North (~70 000 km^2) covers modern Bryansk, Kaluga, Orel, and western Tula Oblasts, comprising Bryansk, Starodub, Trubchevsk, Vshchizh, Karachev, Novosil, Tarusa, Mtsensk, Odoev, Vorotynsk, Belyov, and Kozelsk. Following the 1239AD sack of southern Chernigov, Prince Roman Mikhailovich transferred the senior Chernigov-Bryansk throne northward into the Bryansk forest (*Debryansk*), triggering a durable demographic shift toward the wooded Upper Oka (*Verchovskie*) principalities. To the south-east, the Kursk-Seversk Steppe Frontier (~48 000 km^2) covers modern Kursk and western Lipetsk and Belgorod Oblasts, comprising Kursk, Rylsk, and Yelets along the Tuskar, Upper Seym, Svapa, and Upper Psyol rivers.

Table 3 demonstrates how the four devolved sub-regions of the Southern Core (Chernigov Core, Novgorod-Seversk, Putivl-Vorgol-Hlukhiv, and Lower Sozh and Gomel) combine with the northern Upper Oka and Kursk-Rylsk appanages to reconstruct the complete demographic trajectory of the wider Chernigov-Seversk lands (800AD - 1300AD) without boundary overlap.

Sub-Region / Devolved Principality Division	800AD	1000AD	1100AD	1200AD	1250AD	1300AD
Chernigov Core (Desna-Snov-Oster)	100 000	150 000	195 000	240 000	100 000	130 000
Novgorod-Seversk (Middle Desna)	30 000	42 000	55 000	70 000	35 000	42 000
Putivl, Vorgol & Hlukhiv (Posemye)	26 000	38 000	50 000	70 000	35 000	38 000
Lower Sozh & Dnieper (Gomel Corridor)	34 000	50 000	65 000	80 000	40 000	50 000
Southern Core Subtotal (72 235 km2)	190 000	280 000	365 000	460 000	210 000	260 000
Bryansk, Starodub & Upper Oka North	90 000	130 000	170 000	210 000	120 000	140 000
Kursk & Rylsk Posemye East	100 000	150 000	200 000	290 000	100 000	120 000
Northern Appanages Subtotal (118 000 km2)	190 000	280 000	370 000	500 000	220 000	260 000
Grand Principality of Chernigov-Seversk Total	380 000	560 000	735 000	960 000	430 000	520 000

Table 3. Sub-regional population trajectories across the southern Chernigov-Seversk principalities and northern Russian appanages (800 - 1300AD).

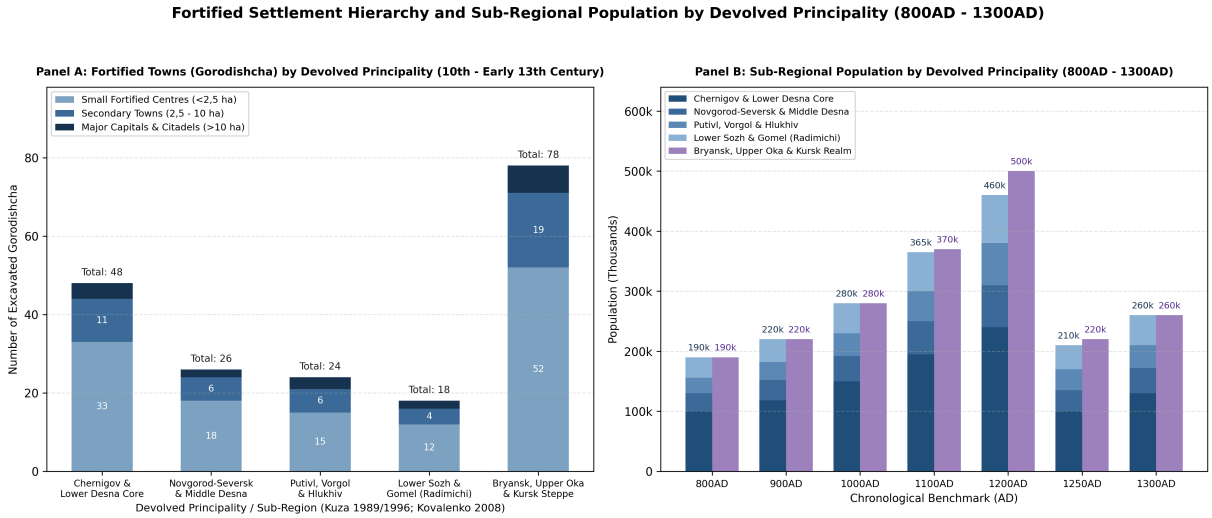


Figure 2. Fortified settlement hierarchy by rampart area and sub-regional population trajectories across the Chernigov-Seversk principalities (800 - 1300AD).

Prehistoric and Early Slavic Settlement (10000BC - 1000AD)

Upper Palaeolithic Mammoth-Hunter Optimum to Bronze Age (10000BC - 1000BC)

The middle Desna incised chalk gorge between Pushkari, Novgorod-Seversk, and Mezin (Mizyn) formed one of the premier Upper Palaeolithic periglacial refugia in Eurasia [9][10][11]. At Mezin (51,83°N, 33,08°E), I. G. Shovkoplyas excavated five circular mammoth-bone *yaranga*-type dwellings enclosing hearths fuel-fed by bone ash, marine amber ornaments transported over 600 km, and geometric meander-engraved ivory bracelets dated to the late Pleniglacial and Bølling-Allerød transition (13000BC - 10000BC). Directly upstream along the Desna bluffs, Pushkari I (Pogon), Chulatovo I-II, Buzhany, and the Novgorod-Seversky rock-shelter document multi-family winter aggregation camps of 60 to 120 inhabitants each. Across the 72 235 km^2 regional core, the 10000BC Late Palaeolithic population is calibrated at 4 500 inhabitants (2 500 - 7 500, 0,06 inh/km^2).

During the Holocene climatic optimum (6000BC - 3000BC), Mesolithic hunter-fishers of the Desna-Sozh tradition transitioned into the Dnieper-Donets II and Pit-Comb Ware Neolithic cultures, occupying low sandy river terraces along the Desna, Snov, Seym, and Sozh (9 000 inhabitants at 6000BC, rising to 18 000 by 4000BC). In the Middle and Late Bronze Age (2000BC - 1000BC), the Sosnytsia variant of the Trzciniec-Komariv cultural horizon and the Marianivka culture along the Seym introduced mixed millet-barley cultivation and floodplain cattle pastoralism, raising the regional population to 52 000 at 1500BC (0,72 inh/km^2) [12].

Yukhnovo, Milograd, Kiev and Kolochin Horizons (1000BC - 700AD)

In the Early Iron Age (700BC - 200BC), the region was divided between two fortified hillfort cultures that established the enduring promontory settlement network of the Desna and Sozh basins [12][13]. The Yukhnovo Culture (named after the Yukhnovo type-hillfort near Novgorod-Seversk) occupied the Middle Desna and Seym valleys, constructing timber-laced earthen ramparts on high right-bank chalk headlands (0,5 - 2,5 ha per hillfort, housing 60 - 150 inhabitants at sites such as Kurove-Yukhnovo, Khorobor, Ponornytsia, and Novgorod-Seversk). To the west, the Milograd Culture (named after the Milograd hillfort

on the Dnieper south of Rechitsa) occupied the Lower Sozh, Braginka, and Dnieper-Desna interfluve (Gomel, Khalch, Milograd, Lyubech, and Gorval), while Scythian forest-steppe farmers fortified the Upper Sula, Psyol, and Vorskla valleys (Vovkivtsi, Pustoviitivka, Kuzemyn, and Romny). By 500BC, regional population reached 85 000 (1,18 inh/km^2).

During the Late La Tène and Roman Iron Age (200BC - 400AD), Late Zarubintsy (Pochep-Harivka horizon, 105 000 at 0AD) and Kiev culture (200AD - 450AD, excavated at Roishche, Desna-Harivka, and Ulianyky) unfortified riverside hamlets proliferated along the Desna, Snov, and Lower Sozh, while the northernmost Chernyakhov culture settlements reached the Sula, Psyol, and southern Seym, driving a Roman-era demographic peak of 140 000 inhabitants at 300AD (1,94 inh/km^2) [14][15]. Following the Hunnic collapse of the Chernyakhov frontier in the late fourth century, the early Slavic Kolochin culture (450AD - 700AD, named after the Kolochin I promontory hillfort near Rechitsa) consolidated along the wooded Dnieper-Sozh-Desna valleys, supporting 115 000 inhabitants at 500AD (1,59 inh/km^2) [15][16].

Volyntsevo, Romny-Severian Expansion and Viking-Age Shestovytsia (700AD - 1000AD)

In the eighth and ninth centuries, Slavic demographic growth accelerated dramatically under the Volyntsevo culture (700AD - 820AD, named after the type-site near Putivl, alongside the fortified Khazar-Severian administrative citadel at Bytytsia on the Psyol) and its direct successor, the Romny-Borshchevo culture (800AD - 1000AD, named after the *Monastyryshche* hillfort at Romny on the Sula) [13][17][18]. Identified with the chronicle Severians (*Severyane*) on the Desna, Seym, and Sula and the southern Radimichi on the Lower Sozh, the Romny population combined fortified promontory refuges (Novotroitske, Gornal, Putivl, Vyr, Romny, Chernigov, Snovsk, and Gomel) with dense clusters of open *selyshcha* equipped with sunken-featured dwellings (*poluzemlyanki*) and domed clay ovens. Archaeological surveys by O. V. Sukhobokov document over 320 Volyntsevo and Romny settlements across the Southern Core, supporting 165 000 inhabitants by 700AD, 190 000 by 800AD, and 220 000 by 900AD.

Simultaneously, the Lower Desna and Dnieper-Sozh corridor became a vital artery of trans-continental Varangian-Rus trade and military organisation [19][20][21]. At Shestovytsia (Korovel, 18 km south-west of Chernigov), excavations by P. Smolychev, D. I. Blifeld, V. P. Kovalenko, and O. P. Motsia revealed a 16 ha fortified hillfort and harbour suburb accompanied by five barrow cemeteries (exceeding 400 mounds) containing tenth-century Scandinavian chamber graves with Frankish swords, warhorses, folding balances, and Abbasid and Samanid silver dirhams (2 400 inhabitants in 900AD, peaking at 3 200 in 1000AD). In the Chernigov retinue necropolises (Chorna Mohyla, Hulbyshche, and Boldyni Hory), B. A. Rybakov's analysis of the monumental *Chorna Mohyla* princely tumulus (40 m diameter, 11 m height, c. 960AD) and the more than 230 barrows of Boldyni Hory demonstrates that tenth-century Chernigov already functioned as the seat of an autonomous princely and boyar *druzhina* (9 500 inhabitants in 900AD, rising to 18 000 by 1000AD). Further west, Lyubech (51,70°N, 30,66°E) and Mokhov (52,02°N, 30,73°E) guarded the Dnieper-Sozh-Desna confluence; Lyubech is listed by Constantine Porphyrogenitus (*De Administrando Imperio*, c. 948AD) as *Telioutza*, one of the key fortified ship-assembly stations on the route from the Varangians to the Greeks (3 500 at Lyubech and 1 950 at Mokhov in 1000AD).

By 1000AD, following Grand Prince Vladimir Svyatoslavich's integration of the Severian and Radimichi lands (984AD Battle on the Pishchana) and the construction of the Desna-Oster-Sula frontier ramparts (Gorodets Ostersky, Pryluky, Romny, and Vovkivtsi), the Southern Core reached 280 000 inhabitants (3,88 inh/km^2), with an urban population of 37 000 (13,2%).

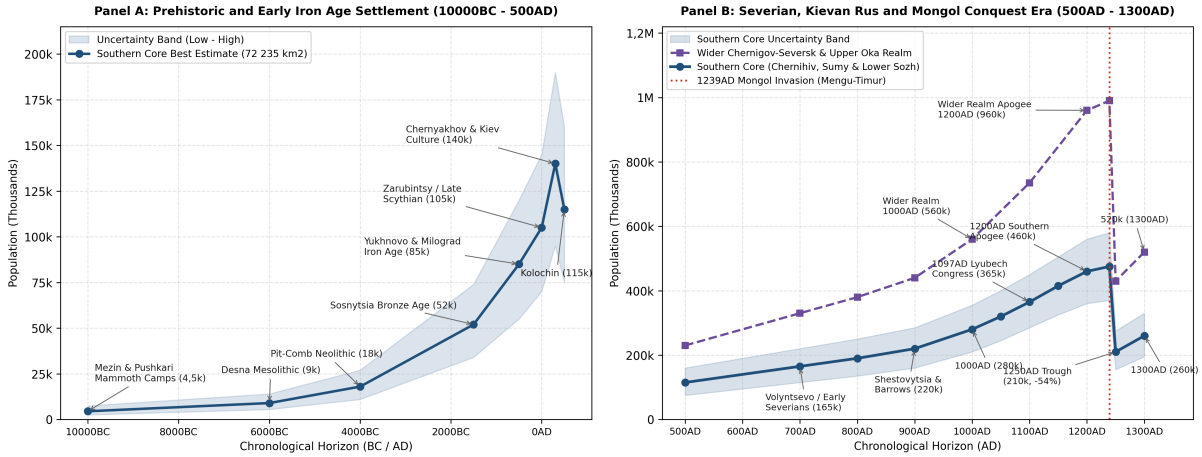


Figure 3. Prehistoric and medieval demographic trajectory of the Left-Bank Desna, Seym, Snov, and Lower Sozh basins from 10000BC to 1300AD.

The Olgovichi Principalities (1000AD - 1239AD)

Dynastic Consolidation from Mstislav Vsevolodovich (1024AD) to the Olgovichi

Following the Battle of Listven (1024AD, fought beside the Listven hillfort 22 km north of Chernigov), Prince Mstislav Vladimirovich established Chernigov as the capital of Left-Bank Rus, founding the stone Transfiguration Cathedral (*Spaso-Preobrazhensky Sobor*, 1030-1036AD) inside the 16 ha *Detinets* [2][19]. Under Svyatoslav Yaroslavich (1054-1076AD) and his son Oleg Svyatoslavich (d. 1115AD), recognised at the Congress of Lyubech (1097AD) as hereditary ruler of the Chernigov-Seversk land, the dynasty fostered intensive internal colonisation across the *Chernigov Opolye*, the middle Desna, the Lower Sozh, and the Posemye steppe frontier [1][3].

Archaeological syntheses by V. P. Kovalenko [4], A. V. Kuza [6][8], P. P. Tolochko [5], Yu. Yu. Morgunov [22], and H. V. Shtykhov [23] catalogue 116 excavated tenth- to early thirteenth-century fortified *gorodishcha* within the Southern Core (48 in the Chernigov Core, 26 in Novgorod-Seversk, 24 in Putivl-Vorgol-Hlukhiv, and 18 in the Lower Sozh and Gomel corridor), plus 78 *gorodishcha* in the northern Russian appanages (194 fortified towns in total across the Grand Principality), surrounded by over 2 100 unfortified rural *selyshcha* (such as the extensively excavated 12 ha rural craft village of Avtunichi).

Urban Archaeology and Fortified Settlement Hierarchy (1200AD)

Applying the standard Rus urban archaeological multipliers calibrated by P. P. Tolochko and A. V. Kuza (120 - 150 inh/ha for densely built *detinets* and *okolny gorod* ramparts; 75 - 100 inh/ha for outer *posad* and *podol* craft quarters, corresponding to 20 - 25 homesteads per hectare at 5,5 - 6,0 persons per household), Table 4 details the reconstructed 1200AD urban hierarchy across the Southern Core and key northern centres.

Urban Centre	Rampart (ha)	Posad (ha)	Total (ha)	1200AD Pop.	1250AD Pop.
Chernigov	160 ha	170 ha	330 ha	35 000	5 000
Novgorod-Seversk	32 ha	43 ha	75 ha	12 000	3 800
Putivl	22 ha	23 ha	45 ha	7 500	3 200

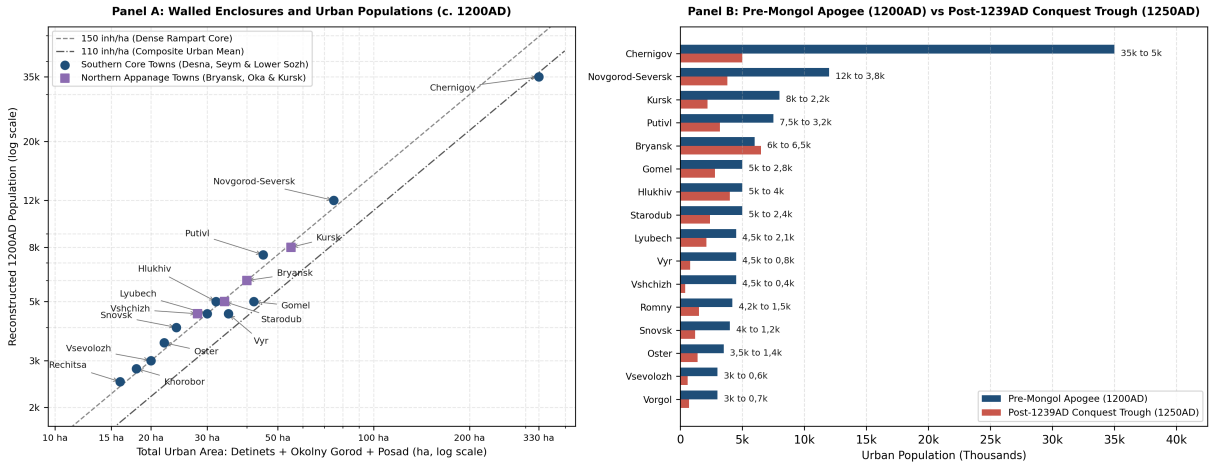


Figure 4. Fortified enclosure areas and pre-Mongol versus post-conquest urban populations across the principal towns of Chernigov-Seversk (1200 - 1250AD).

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Urban Centre	Rampart (ha)	Posad (ha)	Total (ha)	1200AD Pop.	1250AD Pop.
Gomel (Homiy)	18 ha	24 ha	42 ha	5 000	2 800
Hlukhiv	15 ha	17 ha	32 ha	5 000	4 000
Lyubech	14 ha	16 ha	30 ha	4 500	2 100
Vyr	15 ha	20 ha	35 ha	4 500	800
Romny (Romen)	12 ha	16 ha	28 ha	4 200	1 500
Snovsk (Sedniv)	12 ha	12 ha	24 ha	4 000	1 200
Oster	10 ha	12 ha	22 ha	3 500	1 400
Pryluky	10 ha	11 ha	21 ha	3 200	1 100
Vsevolozh	10 ha	10 ha	20 ha	3 000	600
Vorgol	9 ha	11 ha	20 ha	3 000	700
Bragin	9 ha	11 ha	20 ha	3 000	1 400
Khorobor	9 ha	9 ha	18 ha	2 800	900
Unenezh (Nizhyn)	9 ha	9 ha	18 ha	2 800	950
Sosnytsia	9 ha	9 ha	18 ha	2 800	1 150
Rechitsa	8 ha	8 ha	16 ha	2 500	1 500
Kursk (Posemye East)	24 ha	31 ha	55 ha	8 000	2 200
Bryansk (Upper Desna)	20 ha	20 ha	40 ha	6 000	6 500
Starodub (Seversk North)	16 ha	18 ha	34 ha	5 000	2 400
Vshchizh (Upper Desna)	14 ha	14 ha	28 ha	4 500	400

Table 4. Fortified rampart areas, suburban posad extents, and estimated urban populations of major Chernigov-Seversk towns (1200 - 1250AD).

Sub-Regional Synthesis at the 1200AD Apogee (460 000 Inhabitants)

Within the Principality of Chernigov Core (240 000 inhabitants at 1200AD), settlement centred on the fertile *Chernigov Opolye* between the Desna, Snov, and Bilous rivers (12 - 18 inh/km^2), where princely domains (*sela*) such as Orgoshch (2 100), Listven (1 850), Blestovit (2 000), Gurichev (1 800), Snovsk (4 000), and Berezna (2 200) supplied grain, warhorses, and ironwork to the 35 000-strong metropolitan capital of Chernigov (16 ha *Detinets*, 40 ha *Okolny Gorod*, 20 ha *Tretyak*, 85 ha walled *Peredgorodier*, and 170 ha *Podol* and *Leskovitsa* craft suburbs, adorned by five twelfth- and early thirteenth-century brick cathedrals: the Transfiguration, Boris and Gleb, Dormition at Yeletsky Monastery, St Elias, and St Paraskeva Pyatnitsa) [4][19]. Along the Dnieper, middle Desna, and Oster-Udai corridors, fortified *volost* centres - Lyubech (4 500, where B. A. Rybakov excavated the timber-and-earth princely castle of *Zamkova Hora*), Oster (3 500), Pryluky (3 200), Vsevolozh (3 000), Sosnytsia (2 800), Khorobor (2 800), Unenezh (Nizhyn, 2 800), Baturin (2 600), Bakhmach (2 400), Moroviisk (2 200), Varva (2 100), and Lutava (1 800) - anchored an urban network of 68 000 townspeople (28,3% of the core's population) and 172 000 rural inhabitants [1].

In the Principality of Novgorod-Seversk along the Middle Desna and Ivot-Esman tributaries (70 000 inhabitants at 1200AD), administration was ruled from the high chalk headland of Novgorod-Seversk (12 000 inhabitants across 75 ha, comprising the 2,5 ha *Zamkova Hora* citadel, the 30 ha *Okolny Gorod*, and the stone Spaso-Preobrazhensky princely monastery). Immortalised as the seat of Prince Igor Svyatoslavich (1151-1202AD) in the *Tale of Igor's Campaign* (1185AD), this principality controlled the middle Desna crossings and the wooded Ivot, Znob, and Esman valleys, including Voronizh (2 400), Krolevets-Retezh (1 100), Znob (950), Shalyhyne-Obesta (850), Ivot (780), Yampil (740), and Kurove-Yukhnovo (720) [3][4].

Across the Principalities of Putivl, Vorgol, and Hlukhiv in the Posemye and Upper Sula and Psyol valleys (70 000 inhabitants at 1200AD), a chain of fortresses guarded the southern forest-steppe frontier against the Polovtsians (*Cumans*). Putivl (7 500 inhabitants across 45 ha, with a twelfth-century brick princely church excavated on *Nikolske Gorka*) formed a fortified network with Vyr (4 500 across 35 ha), Hlukhiv (5 000 across 32 ha), Romny (4 200 across 28 ha), and Vorgol (3 000 across 20 ha, seat of an autonomous appanage on the Kleven cliffs between 1203AD and 1241AD) [18][22]. Secondary steppe-watch fortresses along the Seym, Tern, Sula, Psyol, and Vorskla included Lipen (Sumy, 2 600), Okhtyrka (2 500), Hlinsk (2 300), Popash (2 200), Konotop (2 100), Viakhan (2 100), Zartyi (1 900), Lipovets (1 850), Nedryhailiv (1 350), and Igoreve Seltsye (1 250, the fortified princely estate with wine and honey cellars looted in the 1146AD chronicle).

In the Lower Sozh and Dnieper-Braginka Corridor of eastern Gomel (80 000 inhabitants at 1200AD), which incorporated the southern Radimichi and Dregovichi-Severian contact zone under Chernigov sovereignty, the riverine settlement network was anchored by Gomel (*Homiy*, 5 000 inhabitants across 42 ha, with weapon-smithing workshops excavated by O. A. Makushnykov), Bragin (3 000), Rechitsa (2 500), Chechersk (2 300), Loyev (2 200), Khalch (1 450), Streshin (1 250), Gorval (1 150), and Zaspa (1 050) [23][24].

Mongol Invasion and Post-Conquest Era (1239AD - 1300AD)

Archaeological Destruction Horizon of October 1239AD

In the autumn of 1239AD, following the devastation of Ryazan (1237AD), Vladimir-Suzdal (1238AD), and Pereyaslavl-Yuzhny (March 1239AD), a Mongol army commanded by Batu's cousin Mengu-Timur (Möngke) and Kadan advanced up the Seym and Desna valleys against the Chernigov-Seversk principalities [3][4]. Archaeological excavations across more than forty *gorodishcha* in the Southern Core reveal a synchronous mid-thirteenth-century destruction horizon marked by vitrified timber-and-earth ramparts, clusters of armour-piercing Mongol tanged arrowheads, hoard depositions of silver *grivnas* and temple rings, and unburied skeletons in collapsed dwellings and church vaults.

Along the Posemye corridor, Putivl, Vyr, Vorgol, Hlukhiv, Popash, and Viakhan were stormed during the initial autumn offensive along the Seym and Kleven in October 1239AD. At Putivl and Vyr, mass graves containing decapitated and arrow-pierced defenders alongside burned artisan quarters confirm the slaughter of the frontier garrisons [18], while Vorgol, Popash, Viakhan, and Zartyi suffered total destruction and permanent urban abandonment after 1239-1241AD. Along the Desna-Snov-Oster approach, Snovsk, Sosnytsia, Khorobor, Vsevolozh, Unenezh, Orgoshch, Listven, and Blestovit were systematically sacked to sever communications before the investment of Chernigov; at Vsevolozh, Blestovit, and Unenezh, burned rampart casemates yielded hundreds of skeletons of defenders, women, and children who perished in the flames [4].

During the siege and sack of Chernigov on 18 October 1239AD, as recorded in the Laurentian and Hypatian Chronicles, Mongol siege engineers surrounded the city and bombarded the walls with heavy counterweight trebuchets (*poroki*) that hurled stones a distance of a bowshot and a half, each stone requiring four men to lift. Despite a desperate relief sortie led outside the ramparts by Prince Mstislav Glebovich of Rylsk and Chernigov - whose troops were routed in open battle - the Mongols breached the *Okolny Gorod* on 18 October 1239AD, put the city to the torch, looted the five brick cathedrals, and slaughtered or enslaved the population [3][4]. Excavations across the 170 ha *Peredgorodier* and *Podol* confirm that the outer suburbs ceased to exist, leaving post-conquest occupation confined to the 16 ha *Detinets* (~5 000 inhabitants in 1250AD).

Demographic Contraction (1250AD) and Post-Mongol Realignment (1240AD - 1300AD)

Between 1239AD and 1250AD, the population of the Southern Core plunged by 54,3%, falling from 460 000 (1200AD) to 210 000 (1250AD, 2,91 inh/km^2), while the urban share dropped from 21,5% to 10,5%. Over 65% of the 116 fortified *gorodishcha* in the Southern Core ceased to function as urban settlements. Following the execution of Grand Prince Mikhail Vsevolodovich of Chernigov and his boyar Fedor at the Golden Horde in September 1246AD for refusing to walk between sacred purification fires [3], the political and demographic geography of the Chernigov-Seversk lands underwent a profound three-way realignment between 1246AD and 1300AD.

Under Mikhail's younger son Roman Mikhailovich the Old (1246-1288AD), the senior princely seat was transferred from devastated Chernigov northward into the dense forests of Bryansk, which expanded from 6 000 (1200AD) to 6 500 (1250AD). Nevertheless, Chernigov, Lyubech (2 500 in 1300AD), and the Lower Sozh towns (Gomel at 3 400, Rechitsa at 1 900, Bragin at 1 750, and Chechersk at 1 650) preserved a reduced episcopal and princely administration, supporting 130 000 inhabitants in the Chernigov Core and 50 000 in the Lower Sozh by 1300AD [24][25].

In 1247AD, Mikhail Vsevolodovich's second son Simeon Mikhailovich established an autonomous hereditary principality at Hlukhiv (1247-1408AD) on the wooded Esman-Kleven plateau (51,68°N, 33,92°E). Sheltered by marshy forests north of the steppe corridor, Hlukhiv retained 4 000 inhabitants in 1250AD and 4 500 in 1300AD, serving as the most resilient urban refuge in the southern Seversk lands until the mid-fourteenth century.

Meanwhile, descendants of the Igorichi line maintained the Principality of Putivl (3 200 inhabitants in 1250AD, recovering to 3 800 by 1300AD) alongside Novgorod-Seversk (3 800 in 1250AD, 4 500 in 1300AD) and Romny (1 950 in 1300AD). Under Golden Horde suzerainty, the Posemye was supervised by the Mongol *baskak* Akhmad, whose creation of tax-exempt *slobody* near Lipovets and Vorgol provoked the 1283-1285AD uprising of Princes Oleg of Rylsk-Vorgol and Svyatoslav of Lipovets recorded in the Laurentian Chronicle [18][25]. By 1300AD, natural recovery across these wooded riverine refugia restored the Southern Core to 260 000 inhabitants (3,60 inh/km^2), while the wider Chernigov-Seversk realm (including the northern Russian appanages) stood at 520 000 inhabitants.

Sources

1. Zaitsev, A. K. *Chernigovskoe Knyazhestvo x-XIII Vv.: Istoriko-Geograficheskoe Issledovanie [the Principality of Chernigov in the 10th-13th Centuries: Historical-Geographical Study]*. Nauka (Institut Istorii SSSR), 1975, https://search.worldcat.org/search?q=Principality+Chernigov+10th+13th+Centuries+Historical+Geographical+Study+Zaitsev+1975.
2. Dimnik, Martin. *The Dynasty of Chernigov, 1054-1146*. Pontifical Institute of Mediaeval Studies, 1994, https://doi.org/10.1017/cbo9780511496479.
3. Dimnik, Martin. *The Dynasty of Chernigov, 1146-1246*. Cambridge University Press, 2003, https://doi.org/10.1017/cbo9780511496479.
4. Kovalenko, V. P. "Urbanizatsiini Protsesy u Chernihovo-Siverskii Zemli x-XIII St. [Urbanisation Processes in the Chernihiv-Siversk Land in the 10th-13th Centuries]." *Arkheolohiya*, vol. 2, 2008, pp. 34-52, https://arheologia.com.ua/index.php/arheologia/issue/archive.
5. Tolochko, P. P. *Drevnerusskiy Feodalnyy Gorod [the Ancient Rus Feudal Town]*. Naukova Dumka, 1989, https://search.worldcat.org/search?q=Ancient+Rus+Feudal+Town+Tolochko+1989.
6. Kuza, A. V. *Malye Goroda Drevnei Rusi [Small Towns of Ancient Rus]*. Nauka, 1989, https://search.worldcat.org/search?q=Small+Towns+Ancient+Rus+Kuza+1989.
7. Tolochko, P. P. "Naselennia Kyivskoi Rusi: Istoryko-Demohrafichnyi Narys [Population of Kievan Rus: An Historical-Demographic Essay]." *Ukrainskyi Istorychnyi Zhurnal*, vol. 3, 2001, pp. 3-18, https://uhj.history.org.ua/.
8. Kuza, A. V. *Drevnerusskie Gorodishcha x-XIII Vv.: Svod Arkheologicheskikh Pamyatnikov [Ancient Rus Hillforts of the 10th-13th Centuries: Corpus of Archaeological Monuments]*. Khristianskoe izdatelstvo, 1996, https://search.worldcat.org/search?q=Ancient+Rus+Hillforts+10th+13th+Centuries+Corpus+Archaeological+Kuza+1996.
9. Shovkoplyas, I. G. *Mezinskaya Stoyanka: K Istorii Srednedneprovskogo Basseyna v Pozdnepaleoliticheskuyu Epokhu [the Mezin Site: On the History of the Middle Dnieper and Desna Basin in the Late Palaeolithic Epoch]*. Naukova Dumka, 1965, https://search.worldcat.org/search?q=Mezin+Site+History+Middle+Dnieper+Desna+Basin+Late+Shovkoplyas+1965.
10. Boriskovsky, P. I. *Paleolit Ukrainy: Istoriko-Arkheologicheskie Ocherki [the Palaeolithic of Ukraine: Historical-Archaeological Essays, Pushkari i and Novgorod-Seversky]*. vol. 40, Izdatelstvo AN SSSR, 1953, https://search.worldcat.org/search?q=Palaeolithic+Ukraine+Historical+Archaeological+Essays+Pushkari+Novgorod+Seversky+Boriskovsky+1953.
11. Velichko, A. A., et al. *Ranniy i Sredniy Paleolit Basseyna Desny [Late Palaeolithic Mammoth-Hunter Settlements of the Desna Basin]*. Institut Geografii RAN, 1997, https://search.worldcat.org/search?q=Late+Palaeolithic+Mammoth+Hunter+Settlements+Desna+Basin+Velichko+1997.
12. Tretyakov, P. N. *Finno-Ugry, Balty i Slavyane Na Dnepre i Volge [Finno-Ugrians, Balts and Slavs on the Dnieper and Volga: Yukhnovo, Milograd and Zarubintsy Horizons]*. Nauka, 1966, https://search.worldcat.org/search?q=Finno+Ugrians+Balts+Slavs+Dnieper+Volga+Yukhnovo+Milograd+Tretyakov+1966.
13. Lyapushkin, I. I. *Dneprovskoe Lesostepnoe Levoberezhe v Epokhu Zheleza [the Dnieper Forest-Steppe Left Bank in the Iron Age and Romny Period]*. vol. 104, Izdatelstvo AN SSSR, 1961, https://search.worldcat.org/search?q=Dnieper+Forest+Steppe+Left+Bank+Iron+Age+Romny+Lyapushkin+1961.
14. Symonovich, E. A., and N. M. Kravchenko. *Pogrebalnye Obryady Plemen Chernyakhovskoy Kultury [Burial Rites of the Chernyakhov Culture Tribes and the Desna-Seym Frontier]*. Nauka, 1983, https://search.worldcat.org/search?q=Burial+Rites+Chernyakhov+Culture+Tribes+Desna+Seym+Frontier+Symonovich+1983.
15. Byrnya, P. P., and R. V. Terpilovsky. *Ranneslavyanskie Poseleniya Dneprovskogo Levoberezhya (Kievskaya i*

Kolochinskaya Kultury) [Early Slavic Settlements of the Dnieper Left Bank]. Naukova Dumka, 1991, https://search.worldcat.org/search?q=Early+Slavic+Settlements+Dnieper+Left+Bank+Byrnya+1991.

16. Baran, V. D. *Davni Sloviany [the Early Slavs of Ukraine]*. Alternatyvy, 1998, https://search.worldcat.org/search?q=Early+Slavs+Ukraine+Baran+1998.
17. Sukhobokov, O. V. *Slavyane Dneprovskogo Levoberezhya (Romenskaya Kultura i Ee Predshestvenniki) [the Slavs of the Dnieper Left Bank: The Romny Culture and Its Predecessors]*. Naukova Dumka, 1975, https://search.worldcat.org/search?q=Slavs+Dnieper+Left+Bank+Romny+Culture+Its+Predecessors+Sukhobokov+1975.
18. Sukhobokov, O. V. *Dniprovske Lisostepove Livoberezhzhia u VIII-XIII St. [The Dnieper Forest-Steppe Left Bank in the 8th-13th Centuries: Putyvl, Vyr, Romny and Posemye]*. Naukova Dumka, 1992, https://search.worldcat.org/search?q=Dnieper+Forest+Steppe+Left+Bank+8th+13th+Centuries+Sukhobokov+1992.
19. Rybakov, B. A. *Drevnosti Chernigova [Antiquities of Chernigov]*. vol. 11, Izdatelstvo AN SSSR, 1949, https://search.worldcat.org/search?q=Antiquities+Chernigov+Rybakov+1949.
20. Motsia, O. P., and A. L. Kazakov. *Davnoruskyi Chernihiv [Ancient Rus Chernihiv and the Shestovytsia Retinue Complex]*. IA NAN Ukrainy / Starodavnii Svit, 2011, https://search.worldcat.org/search?q=Ancient+Rus+Chernihiv+Shestovytsia+Retinue+Complex+Motsia+2011.
21. Androshchuk, Fedir. *Vikings in the East: Essays on Contacts Along the Road to Byzantium (800-1100)*. Uppsala Universitet, 2013, https://doi.org/10.33063/m6c0cy98.
22. Morgunov, Yu. Yu. *Drevnerusskie Pamyatniki Porechya Suly [Ancient Rus Monuments of the Sula and Posemye Frontier]*. Institut Arkheologii RAN, 1996, https://search.worldcat.org/search?q=Ancient+Rus+Monuments+Sula+Posemye+Frontier+Morgunov+1996.
23. Shtykhov, H. V. *Goroda Polotskoy i Chernigovo-Turovskoy Zemel v IX-XIII Vv. [Towns of the Polotsk and Chernigov-Turov Lands in the 9th-13th Centuries]*. Navuka i Tekhnika, 1992, https://search.worldcat.org/search?q=Towns+Polotsk+Chernigov+Turov+Lands+9th+13th+Centuries+Shtykhov+1992.
24. Makushnykov, O. A. *Gomelskae Padnyaprove u v-XIII Stst.: Satsyyalna-Ekanamichnae i Kulturna-Gistarychnae Razvitstse [the Gomel Dnieper-Sozh Region in the 5th-13th Centuries]*. GGU im. F. Skaryny, 2009, https://search.worldcat.org/search?q=Gomel+Dnieper+Sozh+Region+5th+13th+Centuries+Makushnykov+2009.
25. Kuczynski, Stefan Maria. *Ziemie Czernihowsko-Siewierskie Pod Rzadami Litwy [the Chernigov-Seversk Lands in the Post-Mongol and Early Lithuanian Transition]*. Ukraiński Instytut Naukowy, 1936, https://search.worldcat.org/search?q=Chernigov+Seversk+Lands+Post+Mongol+Early+Lithuanian+Transition+Kuczynski+1936.